



Habib University

EE-437 Renewable energy systems – Fall 2025

ID: Solution

Time = 15 minutes

Quiz 04 – CLO 1

Max Points: 20

CLO 1 - 20 marks

Question 1 [2 points]: Why does hydropower have a much larger share of electric power worldwide compared to solar and wind (Any 4 points)?

High utilization

Very high predictability

lowest LCOE

More ^{uses} ~~advantages~~ than just electricity

Question 2 [3 points]: What factors contribute to the efficiency of a pumped storage plant?

Height difference

Turbine/pump efficiency

Motor/generator efficiency

Question 3 [2 points]: What are the negative impacts of building dams?

Impact on fish

Flooding behind dam

Question 4 [2 points]: What is the approximate efficiency of photosynthesis and plant growth?

3-6% photosynthesis

0.25-1% plant growth

Question 5 [2 points]: Give 1 example of each biomass type: ligneous, herbaceous, vegetable oil, starchy.

L: wood

H: Grass

V: Palm

S: wheat

Question 6 [2 points]: Why do biomass thermal plants have lower efficiency than conventional power plants?

Low temperature of combustion

High water content

Question 7 [1 point]: Sort the following in order of lowest to highest energy density: coal, wood, straw.

Straw < Wood < Coal

Question 8 [2 points]: Compare biomass with natural gas.

Bio gas has lower CH_4 , higher H_2O

" " " energy density / heating value

Question 9 [1 point]: Why can't water from a geothermal well be directly used for turbines?

Because of impurities

Question 10 [1 points]: Why are geothermal and heat pumps often talked about together?

BOTH take energy from earth

Heat pumps are more efficient underground

Question 11 [1 point]: What is the formula for coefficient of performance in terms of T_C and T_H ?

$$COP = \frac{T_{cold}}{T_{hot} - T_{cold}}$$

Question 12 [1 point]: Why is nuclear often put in the same category as renewables even though it isn't a renewable energy source?

No CO_2 emissions